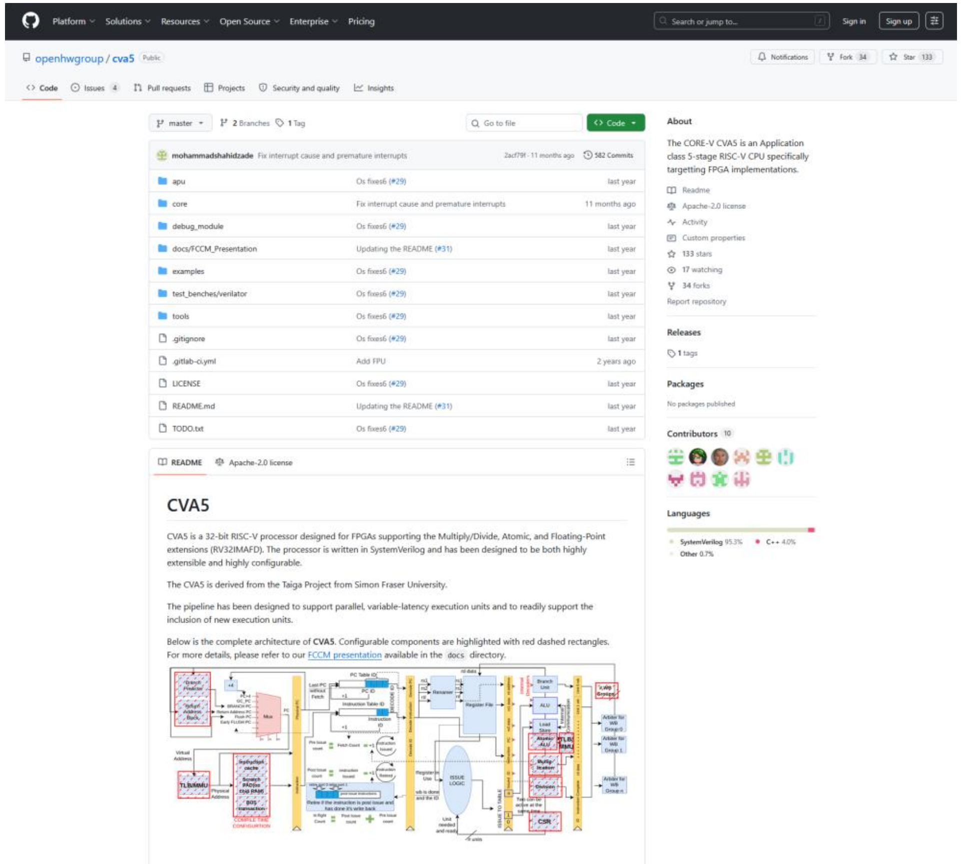
aLean-Tec
adding value to open source HW

Project pitch: evolving CVA5 for TinyAI

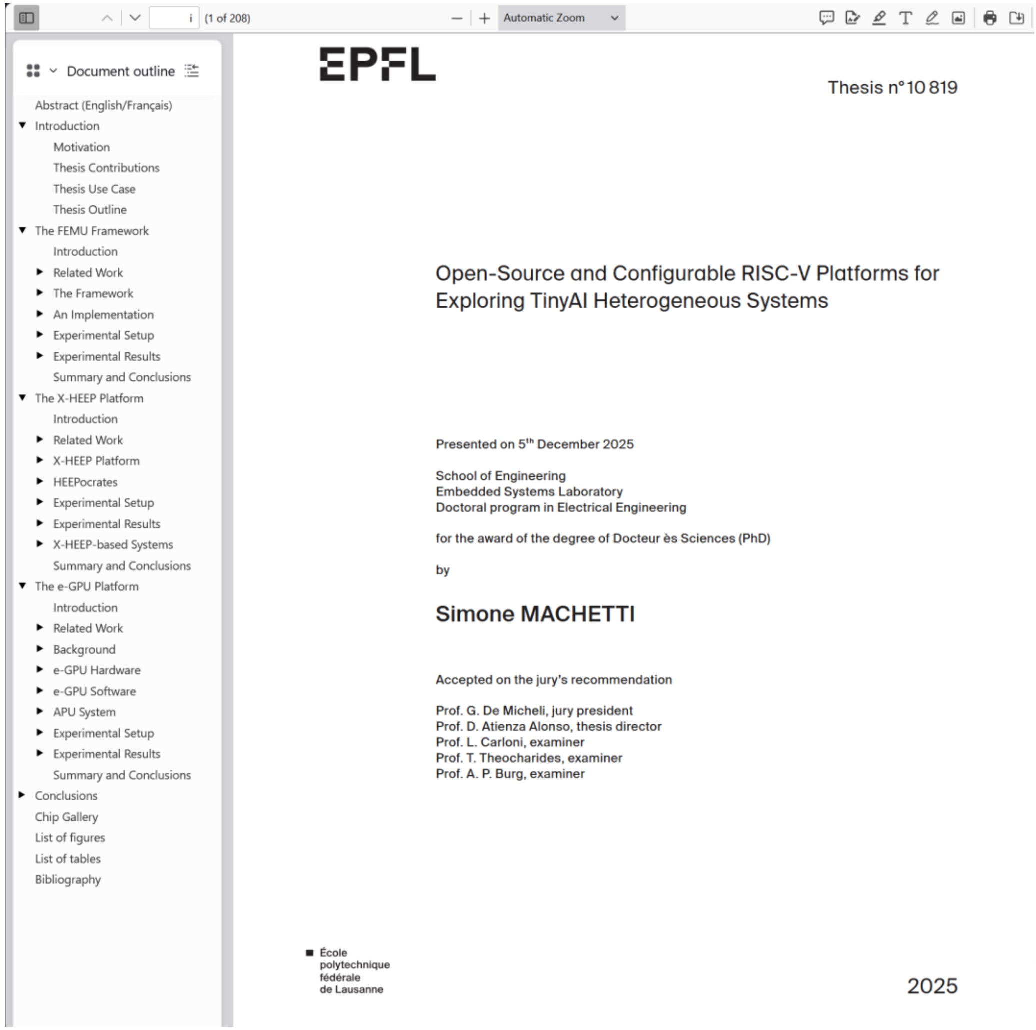
Alfredo Herrera, principal engineer.

Starting point is the merge of:

- The existing project

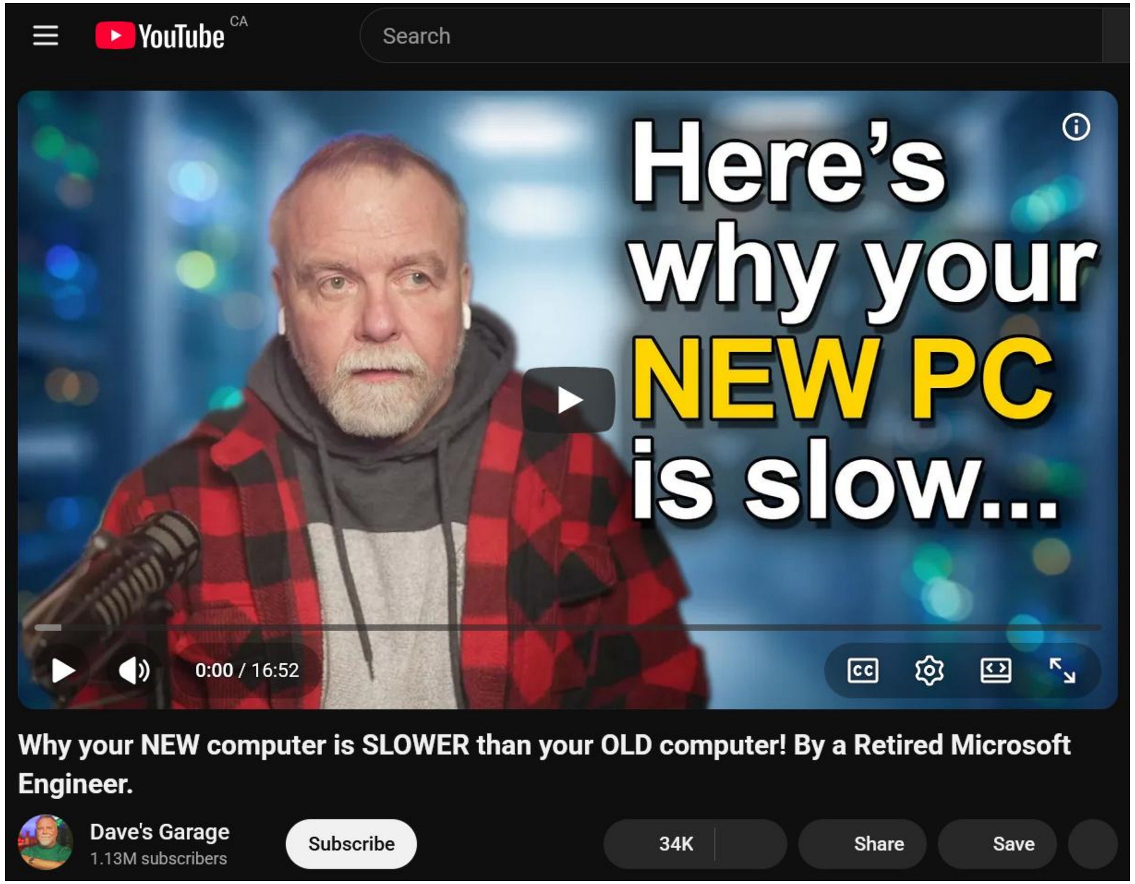


- What's needs to support this PHD thesis



Framework of project

- Goal is to maintain and grow the CVA5 project.
- While sharing understanding of :
  - full system and architecture choices
  - performance measurement and trade-offs
    - memory budget
    - startup time budget
    - disc (bulk storage footprint) budget.
    - CPU usage budget.
  - SW (algorithm) and HW acceleration
  - Multi-thread and Multi-core execution
  - Register maps, boot code and APIs.
  - Security and encryption
  - GPUs and their uses (including AI)



Initial plan

- Q3’2026:
  - Learn to verify and implement in FPGA the current version of the CVA5
  - Prepare demo of project, to present at Ottawa RISC-V user group meeting:
    - Demo will show simulations with Verilator and a commercial simulator
    - Demo will include instructions on running CVA5 SW on QEMU
    - Demo will include CVA5 running on FPGA
- Q4’2026:
  - Focus will be in developing on-boarding and ramp-up process for adopters of CVA5
- Q1’2027:
  - Prepare paper, poster, presentation and demo for a conference (TBD)
- Q2’2027:
  - Register map, bare metal boot code, APIs.
  - Develop board support package (BSP).